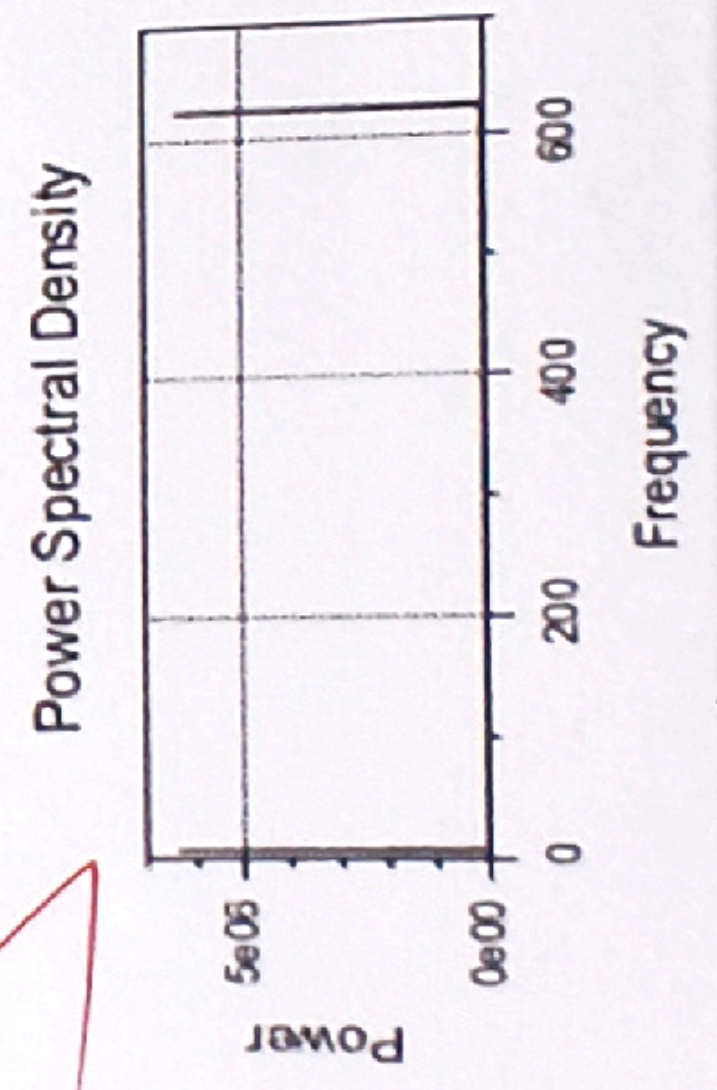
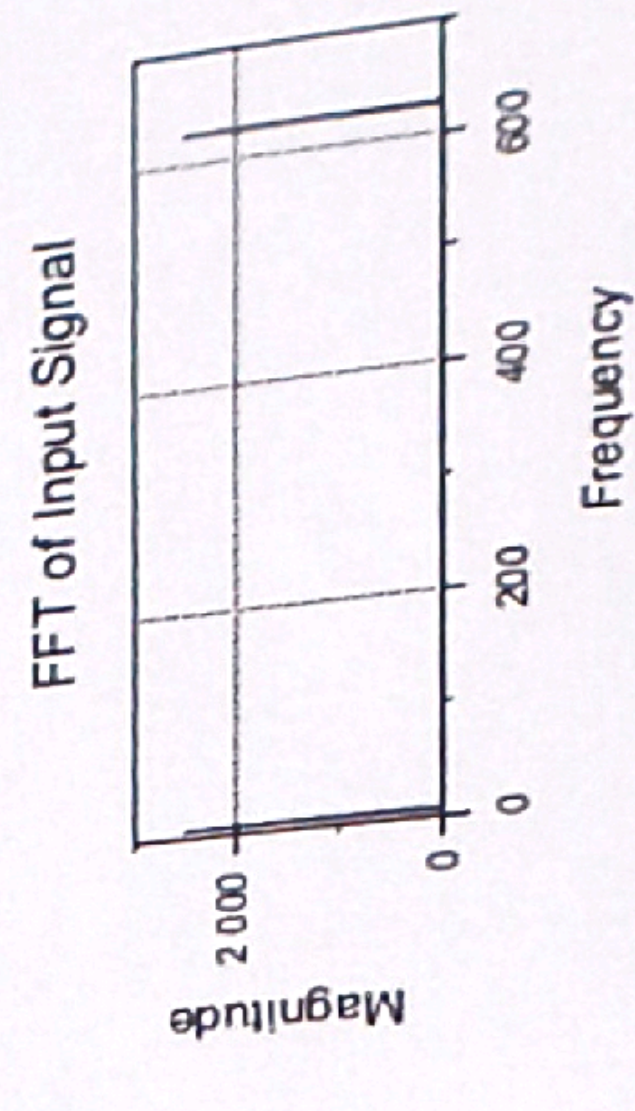
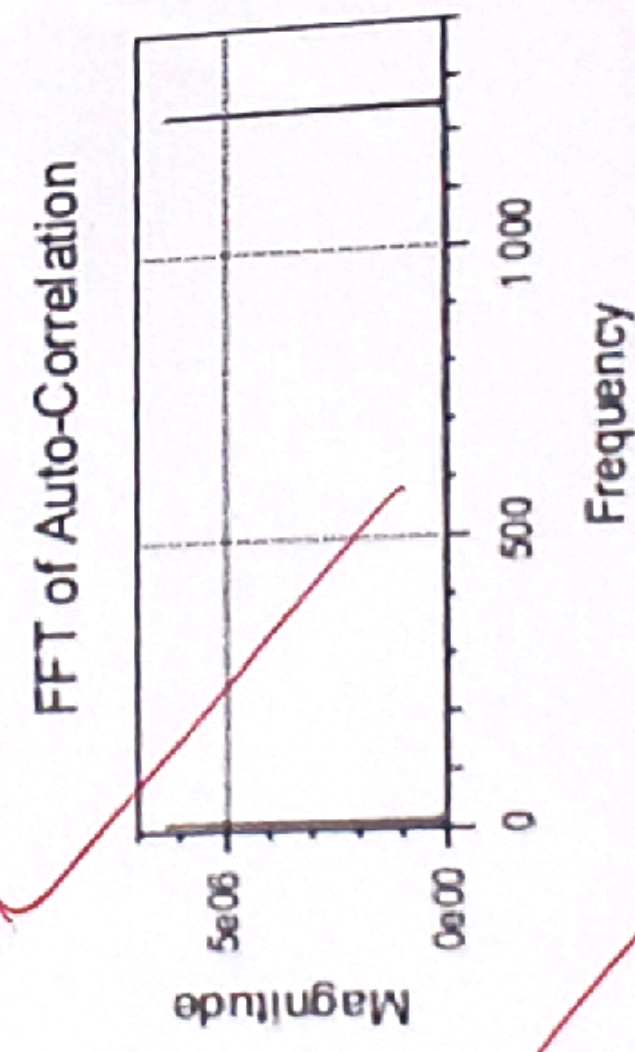
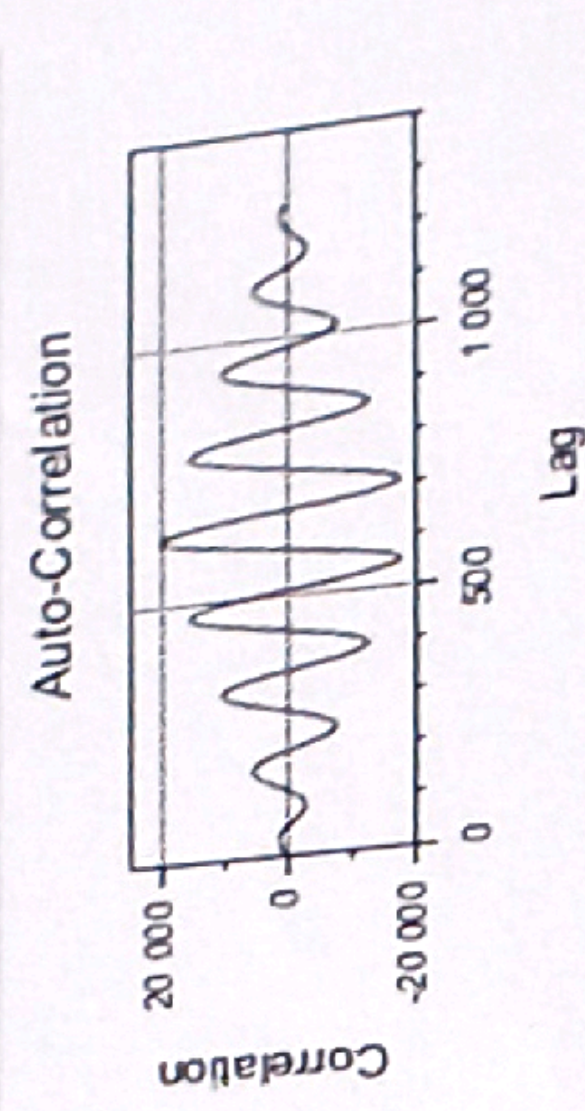
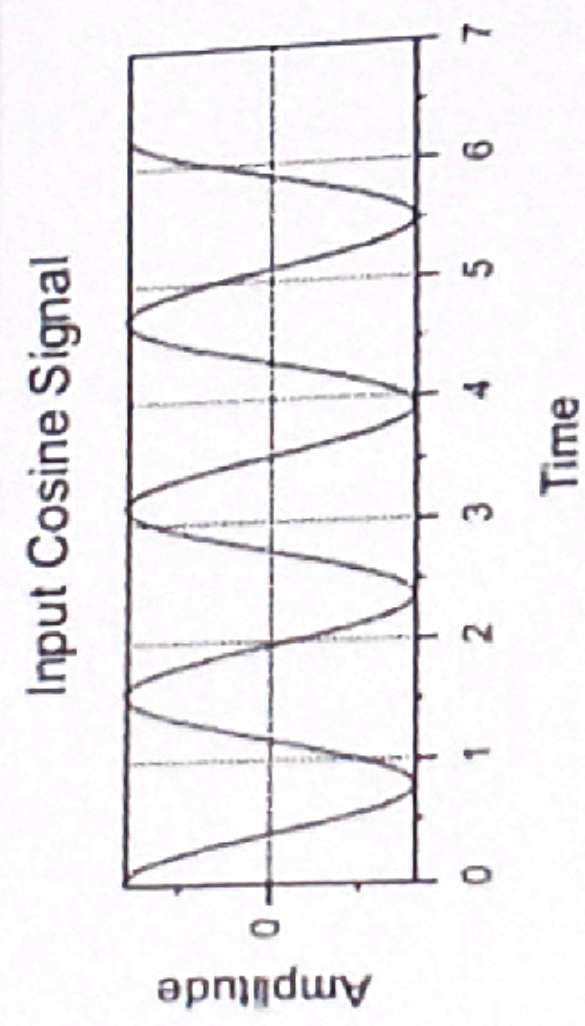




19/05/20

Exp-7. Simulation of Auto-correlation and PSD using Scilab

AIM:

To simulate the auto-correlation function and Power Spectral Density (PSD) of a cosine signal using Scilab.

SOFTWARE REQUIRED:

Scilab Software

CODE:

```
clc;  
clear;  
close;  
  
t = 0:0.01:2*pi;  
x = 8*cos(4*t);  
  
subplot(3,2,1);  
plot(t,x);  
xgrid();  
title("Input Cosine Signal");  
xlabel("Time");  
ylabel("Amplitude");  
  
av = xcorr(x,x);  
subplot(3,2,2);  
plot(av);  
xgrid();  
title("Auto-correlation");  
xlabel("Lag");  
ylabel("Correlation");  
  
V = fft(av);  
subplot(3,2,3);  
plot(abs(V));  
xgrid();
```

```
title ("FFT of Auto-correlation");
```

```
xlabel ("Frequency");
```

```
ylabel ("Magnitude");
```

```
fw = fft(x);
```

```
subplot (3,2,4);
```

```
plot (abs (fw));
```

```
xgrid();
```

```
title ("FFT of Input signal");
```

```
xlabel ("Frequency");
```

```
ylabel ("Magnitude");
```

```
fw2 = (abs (fw)).^2;
```

```
subplot (3,2,5);
```

```
plot (fw2);
```

```
xgrid();
```

```
title ("power Spectral Density");
```

```
xlabel ("Frequency");
```

```
ylabel ("power");
```

RESULT:

Thus, the auto-correlation and PSD of the given cosine signal were successfully obtained using Scilab.

Scilab Program (20)	Waveforms and Tabulation (20)	Calculation and Result (20)	VIVA -VUE (20)	RECORD, OBSERVATION (20)
20	20	20	10	20 90